Predicting the prices of Avacados

About the data-

> The dataset represents weekly 2018 retail scan data for National retail volume (units) and price. Retail scan data comes directly from retailers' cash registers based on actual retail sales of Hass avocados. Starting in 2013, the table below reflects an expanded, multi-outlet retail data set. Multi-outlet reporting includes an aggregation of the following channels: grocery, mass, club, drug, dollar and military. The Average Price (of avocados) in the table reflects a per unit (per avocado) cost, even when multiple units (avocados) are sold in bags. The Product Lookup codes (PLU's) in the table are only for Hass avocados. Other varieties of avocados (e.g. greenskins) are not included in this table.

Some relevant columns in the dataset:

- Date - The date of the observation
- AveragePrice - the average price of a single avocado
- type - conventional or organic
- year - the year
- Region - the city or region of the observation
- Total Volume - Total number of avocados sold
- 4046 - Total number of avocados with PLU 4046 sold
- 4225 - Total number of avocados with PLU 4225 sold
- 4770 - Total number of avocados with PLU 4770 sold

In [1]:
```
#display image using python
from IPython.display import Image
url = 'https://img.etimg.com/thumb/msid-71806721,width-650,imgsize-807917,,resizemo
Image(url,height=300,width=400)
```

Out[1]:

In [2]:
```
#importing libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set()
import warnings
warnings.filterwarnings('ignore')
#importing the dataset
data = pd.read_csv('../input/avocado-prices/avocado.csv',index_col=0)
# Check the data
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 18249 entries, 0 to 11
Data columns (total 13 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Date          18249 non-null  object 
 1   AveragePrice  18249 non-null  float64
 2   Total Volume  18249 non-null  float64
 3   4046          18249 non-null  float64
 4   4225          18249 non-null  float64
 5   4770          18249 non-null  float64
 6   Total Bags    18249 non-null  float64
 7   Small Bags    18249 non-null  float64
 8   Large Bags    18249 non-null  float64
 9   XLarge Bags   18249 non-null  float64
 10  type          18249 non-null  object 
 11  year          18249 non-null  int64  
 12  region        18249 non-null  object 
dtypes: float64(9), int64(1), object(3)
memory usage: 1.9+ MB
```

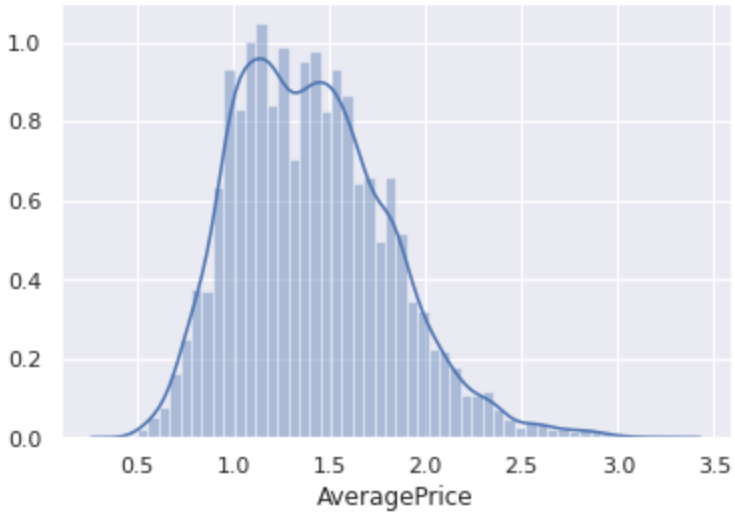
There are 3 categorical features and luckily no missing value. Let's explore the data further.

In [3]:
```
data.head(3)
```

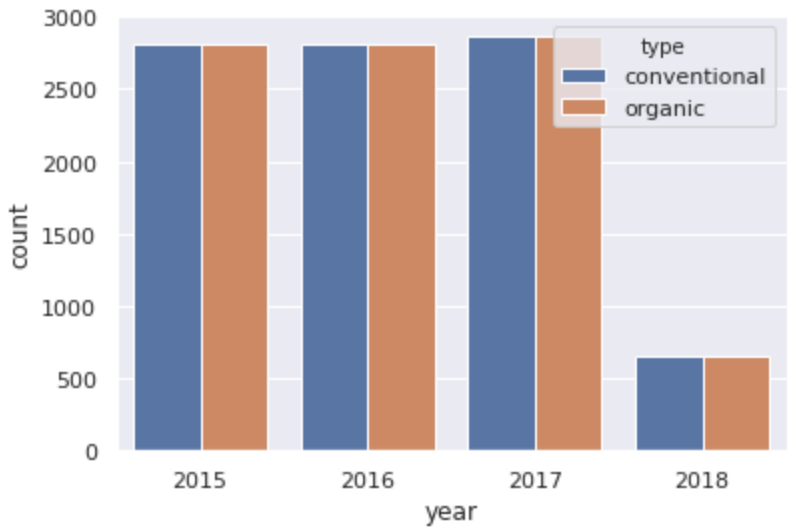
Out[3]:

	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XL
0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	
1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	
2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	

In [4]:
```
sns.distplot(data['AveragePrice']);
```



In [5]:
```
sns.countplot(x='year',data=data,hue='type');
```

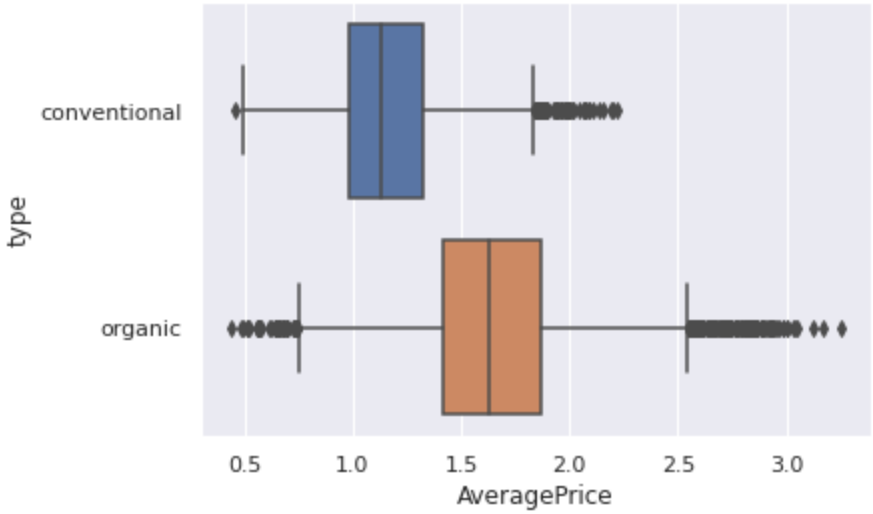


There are almost equal numbers of conventional and organic avacados. Though, there is very less observations in the year 2018.

In [6]:
```
data.year.value_counts()
```

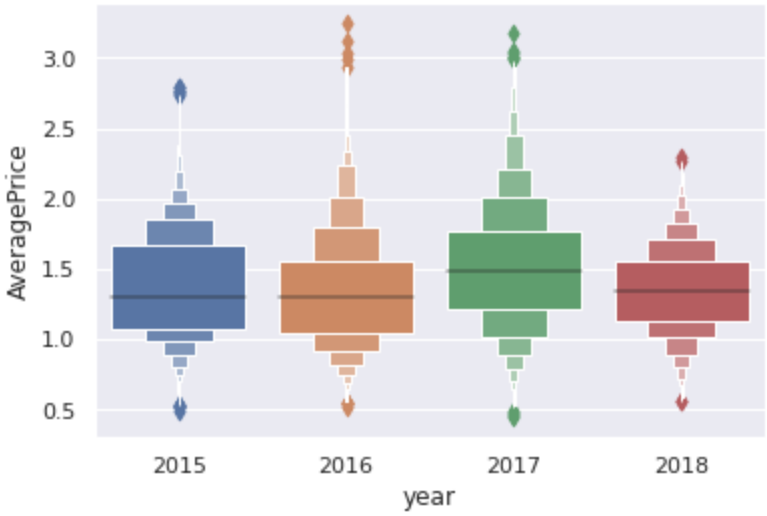
Out[6]:
```
2017    5722
2016    5616
2015    5615
2018    1296
Name: year, dtype: int64
```

In [7]:
```
sns.boxplot(y="type", x="AveragePrice", data=data);
```



Organic avocados are more expensive. This is obvious, because their cultivation is more expensive and we all love natural products and are willing to pay a higher price for them.

In [8]:
```
data.year=data.year.apply(str)
sns.boxenplot(x="year", y="AveragePrice", data=data);
```



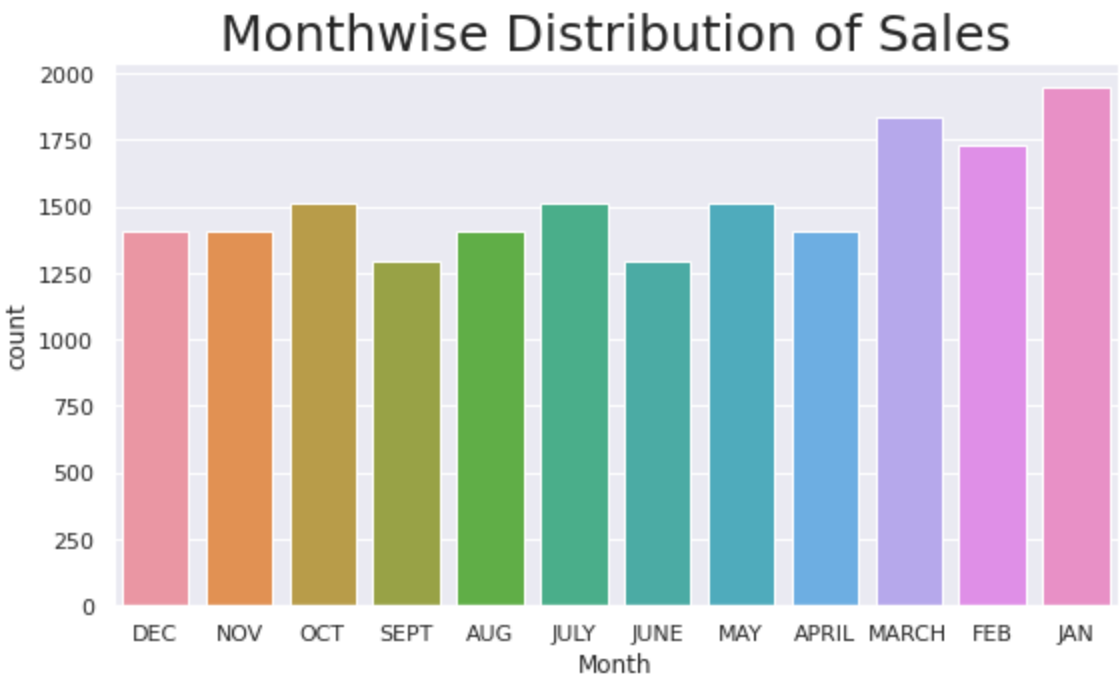
Avacados were slightly more expensive in the year 2017.(as there was shortage due to some reasons)

Dealing with categorical features.

In [9]:
```
data['type']= data['type'].map({'conventional':0,'organic':1})

# Extracting month from date column.
data.Date = data.Date.apply(pd.to_datetime)
data['Month']=data['Date'].apply(lambda x:x.month)
data.drop('Date',axis=1,inplace=True)
data.Month = data.Month.map({1:'JAN',2:'FEB',3:'MARCH',4:'APRIL',5:'MAY',6:'JUNE',7
```

In [10]:
```
plt.figure(figsize=(9,5))
sns.countplot(data['Month'])
plt.title('Monthwise Distribution of Sales',fontdict={'fontsize':25});
```



It implies that sales of avacado see a rise in January, Febuary and March.

Preparing data for ML models

In [11]:
```
# Creating dummy variables
dummies = pd.get_dummies(data[['year','region','Month']],drop_first=True)
df_dummies = pd.concat([data[['Total Volume', '4046', '4225', '4770', 'Total Bags',
       'Small Bags', 'Large Bags', 'XLarge Bags', 'type']],dummies],axis=1)
target = data['AveragePrice']

# Splitting data into training and test set
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df_dummies,target,test_size=0.3

# Standardizing the data
cols_to_std = ['Total Volume', '4046', '4225', '4770', 'Total Bags', 'Small Bags','
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
scaler.fit(X_train[cols_to_std])
```

```
X_train[cols_to_std] = scaler.transform(X_train[cols_to_std])
X_test[cols_to_std] = scaler.transform(X_test[cols_to_std])
```

In [12]:
```
#importing ML models from scikit-learn
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
```

In [13]:
```
#to save time all models can be applied once using for loop
regressors = {
    'Linear Regression' : LinearRegression(),
    'Decision Tree' : DecisionTreeRegressor(),
    'Random Forest' : RandomForestRegressor(),
    'Support Vector Machines' : SVR(gamma=1),
    'K-nearest Neighbors' : KNeighborsRegressor(n_neighbors=1),
    'XGBoost' : XGBRegressor()
}
results=pd.DataFrame(columns=['MAE','MSE','R2-score'])
for method,func in regressors.items():
    model = func.fit(X_train,y_train)
    pred = model.predict(X_test)
    results.loc[method]= [np.round(mean_absolute_error(y_test,pred),3),
                          np.round(mean_squared_error(y_test,pred),3),
                          np.round(r2_score(y_test,pred),3)
                         ]
```

Deep Neural Network

In [14]:
```
# Splitting train set into training and validation sets.
X_train, X_val, y_train, y_val = train_test_split(X_train,y_train,test_size=0.20)

#importing tensorflow libraries
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Activation,Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping

#creating model
model = Sequential()
model.add(Dense(76,activation='relu',kernel_initializer=tf.random_uniform_initializ
    bias_initializer=tf.random_uniform_initializer(minval=-0.1, maxval=0.1)))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initiali
    bias_initializer=tf.random_uniform_initializer(minval=-0.1, maxval=0.1)))
model.add(Dropout(0.5))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initiali
    bias_initializer=tf.random_uniform_initializer(minval=-0.1, maxval=0.1)))
model.add(Dropout(0.5))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initiali
    bias_initializer=tf.random_uniform_initializer(minval=-0.1, maxval=0.1)))
```

```
model.add(Dropout(0.5))
model.add(Dense(1))

model.compile(optimizer='Adam', loss='mean_squared_error')
early_stop = EarlyStopping(monitor='val_loss', mode='min', verbose=0, patience=10)
```

In [15]:
```
model.fit(x=X_train.values,y=y_train.values,
          validation_data=(X_val.values,y_val.values),
          batch_size=100,epochs=150,callbacks=[early_stop])
```

```
Epoch 1/150
103/103 [==============================] - 1s 7ms/step - loss: 0.2831 - val_loss: 0.
0742
Epoch 2/150
103/103 [==============================] - 1s 5ms/step - loss: 0.1099 - val_loss: 0.
0726
Epoch 3/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0954 - val_loss: 0.
0485
Epoch 4/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0830 - val_loss: 0.
0469
Epoch 5/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0751 - val_loss: 0.
0409
Epoch 6/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0700 - val_loss: 0.
0432
Epoch 7/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0650 - val_loss: 0.
0371
Epoch 8/150
103/103 [==============================] - 1s 6ms/step - loss: 0.0605 - val_loss: 0.
0338
Epoch 9/150
103/103 [==============================] - 1s 6ms/step - loss: 0.0569 - val_loss: 0.
0344
Epoch 10/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0535 - val_loss: 0.
0315
Epoch 11/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0507 - val_loss: 0.
0297
Epoch 12/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0485 - val_loss: 0.
0342
Epoch 13/150
103/103 [==============================] - 1s 6ms/step - loss: 0.0475 - val_loss: 0.
0310
Epoch 14/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0477 - val_loss: 0.
0280
Epoch 15/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0456 - val_loss: 0.
0297
Epoch 16/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0445 - val_loss: 0.
0293
Epoch 17/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0433 - val_loss: 0.
0274
Epoch 18/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0433 - val_loss: 0.
0284
Epoch 19/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0404 - val_loss: 0.
```

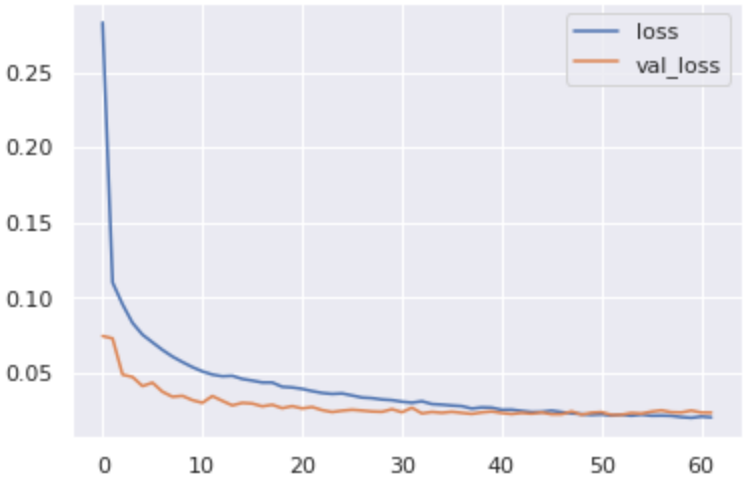
```
0262
Epoch 20/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0400 - val_loss: 0.
0275
Epoch 21/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0390 - val_loss: 0.
0260
Epoch 22/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0375 - val_loss: 0.
0269
Epoch 23/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0363 - val_loss: 0.
0249
Epoch 24/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0357 - val_loss: 0.
0236
Epoch 25/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0360 - val_loss: 0.
0245
Epoch 26/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0346 - val_loss: 0.
0251
Epoch 27/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0332 - val_loss: 0.
0246
Epoch 28/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0328 - val_loss: 0.
0240
Epoch 29/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0319 - val_loss: 0.
0237
Epoch 30/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0315 - val_loss: 0.
0255
Epoch 31/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0305 - val_loss: 0.
0235
Epoch 32/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0298 - val_loss: 0.
0265
Epoch 33/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0308 - val_loss: 0.
0227
Epoch 34/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0289 - val_loss: 0.
0237
Epoch 35/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0285 - val_loss: 0.
0231
Epoch 36/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0279 - val_loss: 0.
0238
Epoch 37/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0275 - val_loss: 0.
0230
Epoch 38/150
```

```
103/103 [==============================] - 0s 5ms/step - loss: 0.0259 - val_loss: 0.
0224
Epoch 39/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0267 - val_loss: 0.
0234
Epoch 40/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0265 - val_loss: 0.
0240
Epoch 41/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0251 - val_loss: 0.
0230
Epoch 42/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0252 - val_loss: 0.
0223
Epoch 43/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0243 - val_loss: 0.
0230
Epoch 44/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0236 - val_loss: 0.
0225
Epoch 45/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0237 - val_loss: 0.
0233
Epoch 46/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0245 - val_loss: 0.
0221
Epoch 47/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0236 - val_loss: 0.
0221
Epoch 48/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0227 - val_loss: 0.
0240
Epoch 49/150
103/103 [==============================] - 1s 6ms/step - loss: 0.0222 - val_loss: 0.
0218
Epoch 50/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0217 - val_loss: 0.
0232
Epoch 51/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0220 - val_loss: 0.
0235
Epoch 52/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0212 - val_loss: 0.
0217
Epoch 53/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0218 - val_loss: 0.
0218
Epoch 54/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0212 - val_loss: 0.
0230
Epoch 55/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0219 - val_loss: 0.
0226
Epoch 56/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0212 - val_loss: 0.
0238
```

```
Epoch 57/150
103/103 [==============================] - 1s 5ms/step - loss: 0.0212 - val_loss: 0.
0247
Epoch 58/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0211 - val_loss: 0.
0234
Epoch 59/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0202 - val_loss: 0.
0233
Epoch 60/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0197 - val_loss: 0.
0246
Epoch 61/150
103/103 [==============================] - 0s 4ms/step - loss: 0.0206 - val_loss: 0.
0232
Epoch 62/150
103/103 [==============================] - 0s 5ms/step - loss: 0.0201 - val_loss: 0.
0232
```

Out[15]: <tensorflow.python.keras.callbacks.History at 0x7f7ea038ae90>

In [16]:
```
losses = pd.DataFrame(model.history.history)
losses[['loss','val_loss']].plot();
```



In [17]:
```
dnn_pred = model.predict(X_test)
```

Results table

In [18]:
```
results.loc['Deep Neural Network']=[mean_absolute_error(y_test,dnn_pred).round(3),m
                                    r2_score(y_test,dnn_pred).round(3)]
results
```

Out[18]:

	MAE	MSE	R2-score
Linear Regression	0.183	0.058	0.636
Decision Tree	0.132	0.040	0.748
Random Forest	0.094	0.018	0.889
Support Vector Machines	0.116	0.027	0.829
K-nearest Neighbors	0.099	0.024	0.852
XGBoost	0.093	0.017	0.895
Deep Neural Network	0.108	0.024	0.853

In [19]:
```
f"10% of mean of target variable is {np.round(0.1 * data.AveragePrice.mean(),3)}"
```

Out[19]:
```
'10% of mean of target variable is 0.141'
```

Let's have a look at methods performing best as they have R2-score close to 1.

In [20]:
```
results.sort_values('R2-score',ascending=False).style.background_gradient(cmap='Gre
```

Out[20]:

	MAE	MSE	R2-score
XGBoost	0.093000	0.017000	0.895000
Random Forest	0.094000	0.018000	0.889000
Deep Neural Network	0.108000	0.024000	0.853000
K-nearest Neighbors	0.099000	0.024000	0.852000
Support Vector Machines	0.116000	0.027000	0.829000
Decision Tree	0.132000	0.040000	0.748000
Linear Regression	0.183000	0.058000	0.636000

Conclusion:

- Except linear regression model, all other models have mean absolute error less than 10% of mean of target variable.
- For this dataset, XGBoost and Random Forest algorithms have shown best results.